Homework 1

⏰ DUE: Friday, 6 November 2026, 23:59 (Türkiye time) on ODTUClass

Released: Fri 23 Oct · Start your 7-day log (Question 1) by Tue 27 Oct · Last day for questions: Wed 4 Nov

The upload closes at exactly 23:59. Aim for 22:00 and enjoy a calm evening.

Read this first 👀

- **Submit one PDF on ODTUClass.** Handwritten (scanned/photographed) or typed answers are both fine, but every calculation must show its steps: the formula, the numbers substituted, and the result. A final number without working gets NO credit.
- Round to 2 decimals unless a question says otherwise. Keep more decimals in intermediate steps.
- **Identify your work.** Write your name, surname and student ID at the top of the first page, and name the file **Name_Surname_StudentID.pdf**
- **Make it gradable.** Answer the questions in order and label every part (1a, 1b, …). Scans must be sharp and upright; a page I cannot read cannot earn points.
- **Use the course conventions**. Compute your results using the methods from lecture. Formulas from other sources (websites, AI tools, other textbooks) may give different answers and will not receive full credit.
- **Collaboration.** You may discuss ideas with classmates, but every number and sentence you submit must be your own. In answering questions, you need to use your own data, ID and graph, so identical answers are easy to spot ☺.
- **Late work.** Late homeworks will NOT BE ACCEPTED.
- **Stuck?** Ask in office hours or on the "Ask a question" forum on our web-site by *Wednesday 4 November.* Questions asked on the due date may not get an answer in time.
- **AI tools (ChatGPT, Gemini, Copilot, …).** You may use them like a tutor, to understand a concept. You may not submit AI-generated answers as your own. This homework is built so that copying an AI answer usually gives wrong or unverifiable results. Bonus asks you to document any AI use; honest documentation is never penalised. *Undisclosed AI-generated work is treated as a violation of METU's academic integrity rules.*

Question	Topic	Points
1	Your own data	30
2	Your personal dataset	20
3	Grade the chatbot	25
4	Misleading graph hunt	25
Bonus	AI-use log and reflection	10
	Total	**110**

✨ Just for fun

Guess-before-you-measure: Before collecting your data for Question 1, write your guess for your daily average on page 1 of your PDF. Afterwards: how close was it to $\bar{x}$? Was it even inside your min–max range?

Questions

Question 1: Your own data (30 points)

Record **one** numerical variable about your own life for **7 consecutive days.** Choose one:

- your daily **phone screen time** in minutes (Screen Time on iPhone, Digital Wellbeing on Android), or
- your daily **number of steps** (your phone's health app).

Attach screenshot(s) that show the daily values, for example the weekly bar charts with values visible. The values in your table must match your screenshots.

a. **[7 pts]** Describe your variable: is it qualitative or quantitative, discrete or continuous, and on which scale of measurement? In this context, what is the experimental unit? Do your 7 values form a population or a sample, and if a sample, of which population? Is the mean of your 7 values a parameter or a statistic? Justify each answer in **one sentence**.

b. **[8 pts]** Make a table with the columns x_i, $x_i - \bar{x}$ and $(x_i - \bar{x})^2$. Then compute the mean, median, mode (if any), range, sample variance, standard deviation and coefficient of variation.

c. **[12 pts]** Find Q_1, Q_3, the IQR, the lower and upper fences and any potential outliers. Draw a box plot **by hand** (a photo is fine) and label the five-number summary on it.

d. **[3 pts]** In 3–4 sentences of your own: which measure of centre best describes your "typical day", and why? Can you explain any unusual value (a weekend, an exam day, a trip…)?

Question 2: Your personal dataset (20 points)

The weekly study hours of 9 STAT 112 students are:

$$x: 4, 7, 7, 9, 10, 12, 13, 15, 31.$$

Let a = the last digit of your student ID and b = (the second-to-last digit of your student ID) + 1. For example, ID 2581347 gives $a = 7$ and $b = 4 + 1 = 5$. Your instructor converts hours into "effort points" with

$$y_i = a + b \cdot x_i$$

a. **[3 pts]** Write down your a and b, and your nine y values.

b. **[6 pts]** Directly from your y values, compute $\bar{y}$, the median, s_y, Q_1, Q_3 and the IQR. Show your work.

c. **[6 pts]** Compute $\bar{x}$, s_x, and the IQR of the original x data. Then use the rules for a linear transformation to check your answers in part (b). Do they agree? If not, find your mistake.

d. **[5 pts]** Is CV_y equal to CV_x? Explain why or why not, and state when they would be equal.

Question 3: Grade the chatbot (25 points)

A student gave this question to an AI chatbot:

> *"Here are last week's daily maximum temperatures (°C).*
> *Ankara: 18, 21, 19, 24, 20, 22, 16.*
> *Antalya: 27, 26, 28, 27, 29, 26, 25.*
> *Summarise and compare the two cities.*
> *Also: Ayşe cycles 30 km to campus at 15 km/h and returns along the same 30 km at 30 km/h.*
> *What is her average speed?"*

The chatbot's answer is shown below, split into numbered statements.

#	Chatbot's statement
S1	Temperature in °C is measured on a ratio scale, so a 24 °C day is twice as warm as a 12 °C day.
S2	Ankara mean: $\bar{x} = (18 + 21 + 19 + 24 + 20 + 22 + 16) / 7 = 140 / 7 = 20$ °C.
S3	Ankara median: the middle (4th) value of the data is 24 °C.
S4	Ankara range: $24 - 16 = 8$ °C.
S5	Ankara variance: $s^2 = \Sigma(x - \bar{x})^2 / n = 42 / 7 = 6$, so $s = 2.45$ °C.
S6	Antalya: $\bar{x} = 26.86$ °C and median = 27 °C. Since the mean is smaller than the median, Antalya's temperatures are clearly right-skewed.
S7	Antalya standard deviation: $s = 1.35$ °C.
S8	To compare the variability of the two cities fairly, we must use the coefficient of variation: CV(Ankara) = 2.45 / 20 = 12.3% and CV(Antalya) = 1.25 / 26.86 = 4.6%. So Ankara is about 2.6 times as variable as Antalya.
S9	Ayşe's average speed is (15 + 30) / 2 = 22.5 km/h.

a. **[18 pts]** Decide for **each** statement whether it is correct or wrong. For every wrong statement, explain the mistake in **one sentence** and give the corrected statement or number.

b. **[7 pts]** Write a corrected comparison of the two cities in 3 sentences: centre, spread, and which city's weather is more predictable.

Question 4: Misleading graph hunt (25 points)

Find a graph published **in the 60 days before the due date (on or after 7 September 2026)** by a Turkish news site, TV channel, newspaper, social-media account, company or public institution (e.g. TÜİK).

a. **[8 pts]** Attach a screenshot of the graph together with its **exact link** and **publication date**. Graphs without a source that I can check get 0 points for the whole question.

b. **[7 pts]** Which variables does the graph show? For each one, give its type (qualitative/quantitative, discrete/continuous) and scale of measurement.

c. **[10 pts]** Name the type of graph. Is it a good choice for these variables? Point out at least one way in which it could mislead a reader (e.g. a truncated axis, 3D effects, a pie chart with too many slices, missing sample size, unequal intervals), or argue convincingly why it does not mislead.

Bonus: AI-use log and reflection (10 points)

Points here are given for honesty and reflection, **not** for whether you used AI.

a. **[6 pts]** If you used any AI tool for this homework: name the tool, paste your prompt(s), and summarise what it answered. For each use, say what you checked yourself and describe one thing it got wrong, left out, or did differently from the STAT 112 rules. If you did not use AI, write "I did not use AI tools for this homework" and describe how you checked one of your answers yourself (for example, recomputing the variance with the shortcut formula, or using the course's step-by-step calculator).

b. **[4 pts]** In 4–5 sentences, describe two **specific** risks of trusting an AI tool with statistics homework, and one good way to use AI as a study tool.